

JEE MAIN REVISION MASTER: STRAIGHT LINES AND PAIR OF STRAIGHT LINES SYLLABUS REDUCED

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 171 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Straight Lines And Pair Of Straight Lines - Core Application	171 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Straight Lines And Pair Of Straight Lines - Core Application (Very High Yield — 171 PYQs)

Core Principle: Key problem pattern in Straight Lines And Pair Of Straight Lines.

■■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "If a straight line drawn through the point of intersection of the lines $4x+3y-1=0$ and $3x+4y-1=0$, meets the co-ordinate axes at the poin..." [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "Let the mid points of the sides of a triangle ABC be $((5/2), 7)$, $((5/2), 3)$ and $(4, 5)$. If its incentre is $(\blacksquare, k)$, then \$..." [Ans: C]
- [2026 Evening Shift] Q#1 (MCQ): "Let $A(1, 2)$ and $C(-3, -6)$ be two diagonally opposite vertices of a rhombus, whose sides AD and BC are parallel to the line $7x-y=14$. If ..." [Ans: B]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 2nd April Morning Shift (Q#1)

Let the mid points of the sides of a triangle ABC be $((5/2), 7)$, $((5/2), 3)$ and $(4, 5)$. If its incentre is (h, k) , then $3h + k$ is equal to :

- (A) 11
- (B) 12
- (C) 13 [Correct]
- (D) 14

Correct Answer: C

Solution Analysis: Let the midpoints of the sides of triangle ABC be $D((5/2), 7)$, $E((5/2), 3)$, $F(4, 5)$. We have to find the incentre (h, k) of triangle ABC. Step 1: Use the fact about the medial triangle. The triangle formed by joining the midpoints of the sides of a triangle is called the medial triangle. If the midpoints are known, then the original triangle can be reconstructed using: $A = E + F - D$, $B = F + D - E$, $C = D + E - F$. Now calculate each vertex. Vertex A: $A = ((5/2), 3) + (4, 5) - ((5/2), 7)$
 $A = (4, 1)$ Vertex B: $B = (4, 5) + ((5/2), 7) - ((5/2), 3)$ $B = (4, 9)$ Vertex C: $C = (\frac{5}{2}, 7) + (\frac{5}{2}, 3) - (4, 5)$